

EC400 – Static Optimization and Fixed Points

Problem Set 1: Solutions

1. Show that the general quadratic form

$$a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2$$

can be written as

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix};$$

and find its unique symmetric representation.

Solution: Multiply the vector and the matrices and show that it yields the quadratic form. The symmetric representation is

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12}/2 \\ a_{12}/2 & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

2. List all the principal minors of a general (3×3) matrix and denote which are the three leading principal minors.

Solution: For a general (3×3) matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

There is one 3rd order principal minor, namely $\det(A)$.

There are three 2nd order principal minors:

- (a) $\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ formed by deleting column 3 and row 3 from A ;
- (b) $\det \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix}$ formed by deleting column 2 and row 2 from A ;
- (c) $\det \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}$ deleting column 1 and row 1 from A .

Then, there are three 1st order principal minors:

- (a) $\det(a_{11})$ formed by deleting columns and rows 2,3;

- (b) $\det(a_{22})$ formed by deleting columns and rows 1,3;
 (c) $\det(a_{33})$ formed by deleting columns and rows 1,2.

Finally, the leading principal minors are $\det(a_{11})$, $\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, and $\det(A)$.

3. Determine the definiteness of the following symmetric matrices:

$$(a) \begin{pmatrix} 0 & 0 \\ 0 & c \end{pmatrix}; \quad (b) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}; \quad (c) \begin{pmatrix} -3 & 4 \\ 4 & -6 \end{pmatrix}; \quad (d) \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 5 \\ 0 & 5 & 6 \end{pmatrix}.$$

Solution: (a) Note that $\det(M_{22}) = 0$ and that $\det(A) = 0$. The definiteness of A depends completely on the sign of c : the matrix is positive semidefinite if $c \geq 0$ and negative semidefinite if $c \leq 0$. This is obvious when we look at the corresponding quadratic form cx_2^2 . (b) Positive definite. (c) Negative definite; (d) Indefinite.

4. Consider the following quadratic form

$$Q(\mathbf{x}) = ax_1^2 + bx_2^2 + 2abx_1x_2.$$

For what values of the parameter values, a and b , is the quadratic form $Q(\mathbf{x})$ indefinite? Plot your answer in $\mathbb{R}^2$.

Solution: $Q(x)$ is indefinite if it is neither positive semidefinite nor negative semidefinite. $Q(x)$ is positive semidefinite is all principal minors are non-negative which requires:

$$a \geq 0, \quad b \geq 0, \quad ab(1 - ab) \geq 0.$$

For these to hold at once it must be that

$$a \geq 0, \quad b \geq 0, \quad 1 \geq ab.$$

$Q(x)$ is negative semidefinite is all principal minors suitably alternate in sign which requires:

$$a \leq 0, \quad b \leq 0, \quad ab(1 - ab) \geq 0.$$

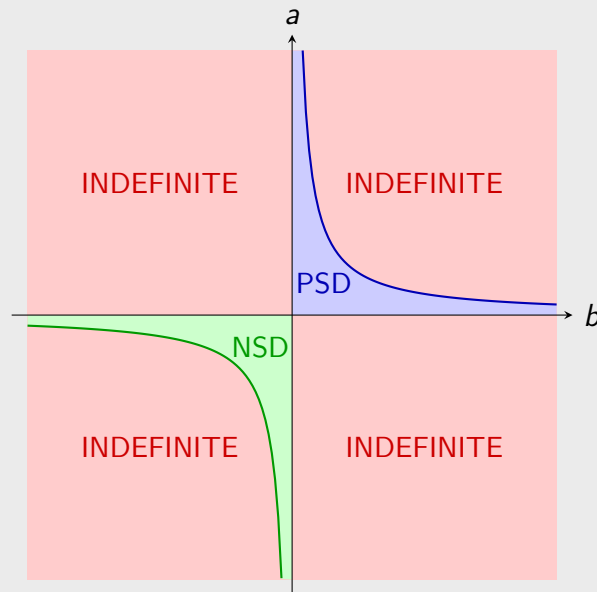
For these to hold at once it must be that

$$a \leq 0, \quad b \leq 0, \quad 1 \geq ab.$$

Thus the quadratic form is indefinite if and only if

$$ab > 1 \quad \text{or} \quad ab < 0.$$

The plot below depicts the answer:



5. Approximate e^x at $x = 0$ with a Taylor polynomial of order three and four. Then compute the values of these approximation at $h = 0.2$ and at $h = 1$ and compare with the actual values.

Solution: Taylor approximations amount to:

$$e^h \approx 1 + h + \frac{1}{2}h^2 + \frac{1}{6}h^3 = P_3(h),$$

$$e^h \approx 1 + h + \frac{1}{2}h^2 + \frac{1}{6}h^3 + \frac{1}{24}h^4 = P_4(h).$$

$$P_3(0.2) = 1.22133...; P_4(0.2) = 1.2214...; e^{0.2} = 1.22140....$$

$$P_3(1) = 2.666...; P_4(1) = 2.70833...; e^1 = 2.71828....$$

6. For each of the following functions on $\mathbb{R}$, determine whether they are quasiconcave, quasiconvex, both, or neither:

(a) e^x ; (b) $\ln x$; (c) $x^3 - x$

Solution: (a) Both, since the function is strictly increasing on $\mathbb{R}$. (b) Both, as before. (c) Neither, graph the function and observe that it has a strict local min and a strict interior local max which implies that C_k^+ and C_k^- are not convex for all k .

7. Let f be a function defined on a convex set U in $\mathbb{R}^n$. In lecture, we have shown that f is a quasiconcave function on U if and only if for all $\mathbf{x}, \mathbf{y} \in U$ and $t \in [0, 1]$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}.$$

State the corresponding theorem for quasiconvex functions and prove it.

Solution: Let f be a function defined on a convex set U in $\mathbb{R}^n$. Then, f is a quasiconvex function on U if and only if for all $\mathbf{x}, \mathbf{y} \in U$ and $t \in [0, 1]$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq \max\{f(\mathbf{x}), f(\mathbf{y})\}.$$

Take any $\mathbf{x}, \mathbf{y} \in U$ and let $\max\{f(\mathbf{x}), f(\mathbf{y})\} = k$. The latter implies that $\mathbf{x}, \mathbf{y} \in C_k^-$. But if f is quasiconvex, C_k^- must be convex and thus $t\mathbf{x} + (1-t)\mathbf{y} \in C_k^-$ for all $t \in [0, 1]$. Thus, for any $t \in [0, 1]$,

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq k = \max\{f(\mathbf{x}), f(\mathbf{y})\}.$$

To prove the converse, take any $\mathbf{x}, \mathbf{y} \in C_k^-$ and note that by assumption for all $t \in [0, 1]$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq \max\{f(\mathbf{x}), f(\mathbf{y})\} \leq k.$$

Thus, $t\mathbf{x} + (1-t)\mathbf{y} \in C_k^-$ for all $t \in [0, 1]$ and all $\mathbf{x}, \mathbf{y} \in C_k^-$, meaning that C_k^- is convex for all k , and hence f is quasiconvex.

8. Prove that a increasing transformation of a quasi-concave function is quasi-concave.

Solution: Consider any increasing transformation $g : \mathbb{R} \rightarrow \mathbb{R}$ and the map $h(\cdot) = g(f(\cdot))$. If for all $\mathbf{x}, \mathbf{y} \in U$ and $t \in [0, 1]$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\},$$

then for all $\mathbf{x}, \mathbf{y} \in U$ and $t \in [0, 1]$

$$\begin{aligned} h(t\mathbf{x} + (1-t)\mathbf{y}) &= g(f(t\mathbf{x} + (1-t)\mathbf{y})) \geq g(\min\{f(\mathbf{x}), f(\mathbf{y})\}) = \\ &= \min\{g(f(\mathbf{x})), g(f(\mathbf{y}))\} = \min\{h(\mathbf{x}), h(\mathbf{y})\}. \end{aligned}$$

9. A commonly used production or utility function on $\mathbb{R}_+^2$ is $f(x, y) = xy$. Check whether it is concave or convex using its Hessian. Then, check whether it is quasiconcave.

Solution: The Hessian is simply

$$D^2f(x, y) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and the second order principal minor is -1 . Since this is negative, then D^2F is indefinite, and F is neither concave nor convex. To check quasiconcavity consider the increasing transformation $g(x) = \log x$, and note that

$$h(x, y) = g(f(x, y)) = \log x + \log y.$$

Thus, $h(x, y)$ is strictly concave and thus quasiconcave. But then $g^{-1}(x) = e^x$ is also a strictly increasing transformation and $h(x, y)$ is quasiconcave. Thus,

$$f(x, y) = g^{-1}(h(x, y)) = xy$$

must be quasiconcave.

10. Show that any continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, satisfying

$$x \frac{\partial f(x)}{\partial x} \leq 0,$$

must be quasiconcave.

Solution: First observe that the conditions in the statement implies that the function always increases in the negative orthant, while it decreases on the positive orthant as

$$\begin{aligned} x < 0 & \text{ implies } \frac{\partial f(x)}{\partial x} \geq 0; \\ x > 0 & \text{ implies } \frac{\partial f(x)}{\partial x} \leq 0. \end{aligned}$$

Thus, the function must have a maximum at $x = 0$. Next we show that implies that any non-empty upper-contour set must be an interval. Take a value $k \leq f(0)$. Look for the smallest value $x \leq 0$ such that

$$f(x) = k.$$

If it exists call it $\underline{x}_k$, otherwise set $\underline{x}_k = -\infty$. By definition of $\underline{x}_k$ and continuity, for any $x \leq 0$, we have that

$$f(x) \geq k \text{ if and only if } x \geq \underline{x}_k.$$

Next look for the largest value $x \geq 0$ such that

$$f(x) = k.$$

If it exists call it $\bar{x}_k$, otherwise set $\bar{x}_k = \infty$. By definition of $\bar{x}_k$ and continuity, for any $x \geq 0$, we have that

$$f(x) \geq k \text{ if and only if } x \leq \bar{x}_k.$$

Thus, $C_k^+ = [\underline{x}_k, \bar{x}_k]$ for all k , which is an interval and thus a convex set. So, the function is quasiconcave.

EC400 – Static Optimization and Fixed Points
Problem Set 2: Solutions

1. For each of the following functions find the critical points and find which of these are local maxima and which local minima:

- (a) $x^4 + x^2 - 6xy + 3y^2$
(b) $x^2 - 6xy + 2y^2 + 10x + 2y - 5$
(c) $xy^2 + x^3y - xy$

Which of the critical points are also global maxima or global minima?

Solution: (a) To find critical points, set the gradient to 0,

$$Df(x, y) = (4x^3 + 2x - 6y, 6y - 6x) = 0.$$

The function possesses 3 critical points, $(0, 0)$, $(1, 1)$, $(-1, -1)$. To find local extrema, focus on the Hessian

$$D^2f(x, y) = \begin{pmatrix} 12x^2 + 2 & -6 \\ -6 & 6 \end{pmatrix}.$$

At $(1, 1)$ and $(-1, -1)$, it is positive definite. Thus, the two critical points are strict local minima. At $(0, 0)$, the Hessian is indefinite. So, $(0, 0)$ is neither a local max nor a local min (such points are called a saddle points). The function is unbounded above and thus possesses no global max. To see this, note that

$$\lim_{x \rightarrow \infty} f(x, x) = x^4 - 2x^2 = \infty.$$

Local min $(1, 1)$ and $(-1, -1)$ are instead strict global min. To see this, observe that the function can be rewritten as

$$f(x, y) = 3(x - y)^2 + (x^2 - 2)x^2.$$

The first term, $3(x - y)^2$, is non-negative, while the second term, $(x^2 - 2)x^2$, is minimized when $x^2 = 1$ since

$$\begin{aligned} \frac{d}{dx}(x^2 - 2)x^2 &= 4(x^2 - 1)x > 0 \Leftrightarrow x \in (-1, 0) \cup (1, \infty), \\ \frac{d}{dx}(x^2 - 2)x^2 &= 4(x^2 - 1)x < 0 \Leftrightarrow x \in (-\infty, -1) \cup (0, 1). \end{aligned}$$

Thus, $(1, 1)$ and $(-1, -1)$ are global min.

(b) To find critical points, set the gradient to 0,

$$Df(x, y) = (10 + 2x - 6y, 2 + 4y - 6x) = 0.$$

The function possesses 1 critical point, $(13/7, 16/7)$. To find local extrema, focus on the Hessian

$$D^2f(x, y) = \begin{pmatrix} 2 & -6 \\ -6 & 4 \end{pmatrix}.$$

The Hessian is indefinite everywhere. So, $(13/7, 16/7)$ is neither a local max nor a local min. The function is unbounded above and below. Thus, it possesses no global extrema. To see this, note that

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x, x) &= -3x^2 + 12x - 5 = -\infty, \\ \lim_{x \rightarrow \infty} f(x, 0) &= x^2 + 10x - 5 = \infty. \end{aligned}$$

(c) To find critical points, set the gradient to 0,

$$Df(x, y) = (y(3x^2 + y - 1), x(2y + x^2 - 1)) = 0.$$

The function possesses several critical points: $(0, 0)$, $(1/\sqrt{5}, 2/5)$, $(-1/\sqrt{5}, 2/5)$, $(1, 0)$, $(-1, 0)$, $(0, 1)$. To find local extrema, focus on the Hessian

$$D^2f(x, y) = \begin{pmatrix} 6xy & 2y + 3x^2 - 1 \\ 2y + 3x^2 - 1 & 2x \end{pmatrix}.$$

At $(1/\sqrt{5}, 2/5)$, it is positive definite, and so the point is a strict local min. At $(-1/\sqrt{5}, 2/5)$, it is negative definite, and so the point is a strict local max. At $(0, 0)$, $(1, 0)$, $(-1, 0)$, $(0, 1)$, the Hessian is indefinite. So, all these critical points are saddle points. The function is unbounded above and below. Thus, it possesses no global extrema. To see this, note that

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x, x) &= x^3 + x^4 - x^2 = \infty, \\ \lim_{x \rightarrow \infty} f(x, -x) &= x^3 - x^4 + x^2 = -\infty. \end{aligned}$$

2. Let $S \subset \mathbb{R}^n$ be an open set and $f : S \rightarrow \mathbb{R}$ be a twice continuously differentiable function. Suppose that $Df(x^*) = 0$. State the weakest sufficient conditions that Hessian at the critical points must satisfy for:

- (i) x^* to be a local max;
- (ii) x^* to be a strict local min.

Solution: (i) $D^2f(\mathbf{x}) \leq 0$ for any $\mathbf{x} \in B(\mathbf{x}^*|\varepsilon)$ for some $\varepsilon > 0$. (ii) $D^2f(\mathbf{x}^*) \geq 0$ & $D^2f(\mathbf{x}) > 0$ for any $\mathbf{x} \in \bar{B}(\mathbf{x}^*|\varepsilon)$ for some $\varepsilon > 0$.

3. Check whether $f(x, y) = x^4 + x^2y^2 + y^4 - 3x - 8y$ is concave or convex by using the Hessian.

Solution: The Hessian is

$$\begin{pmatrix} 12x^2 + 2y^2 & 4xy \\ 4xy & 2x^2 + 12y^2 \end{pmatrix}.$$

All the principal minors, $12x^2 + 2y^2$, $2x^2 + 12y^2$ and $24x^4 + 132x^2y^2 + 24y^4$, are non-negative. Thus, f is a convex function on all $\mathbb{R}^n$.

4. Solve the following problem

$$\max_{x,y} [\min\{x, y\} - x^2 - y^2].$$

Solution: Here differentiability fails and we cannot use first order conditions. Let $f(x, y) = \min\{x, y\} - x^2 - y^2$. Suppose without loss that the maximum $(\bar{x}, \bar{y})$ satisfies $\bar{x} > \bar{y}$. If so, observe that a contradiction arises, since $f(\bar{x}, \bar{y}) < f(\bar{z}, \bar{z})$ for $\bar{z} = (\bar{x} + \bar{y})/2$. The latter must hold as

$$\begin{aligned} f(\bar{x}, \bar{y}) &= \min\{\bar{x}, \bar{y}\} - \bar{x}^2 - \bar{y}^2 = \bar{y} - \bar{x}^2 - \bar{y}^2 < \\ &< \bar{z} - \bar{x}^2 - \bar{y}^2 < \bar{z} - 2\bar{z}^2 = \min\{\bar{z}, \bar{z}\} - 2\bar{z}^2 = f(\bar{z}, \bar{z}). \end{aligned}$$

where the first equality and the first inequality hold since $\bar{x} > \bar{z} > \bar{y}$, where the second inequality holds since by concavity of $-x^2$ as

$$-\frac{1}{2}x^2 - \frac{1}{2}y^2 < -\left(\frac{x+y}{2}\right)^2.$$

Thus, the solution must satisfy $\bar{x} = \bar{y}$ and solve

$$\max_x x - 2x^2.$$

Taking first order conditions we find that $\bar{x} = \bar{y} = 1/4$ which is a maximum since the modified objective function is concave.

5. Find the optimal solution for the following program

$$\max_{x,y} x \quad \text{subject to} \quad x^3 + y^2 = 0.$$

Is the Lagrange approach appropriate?

Solution: This problem is designed to illustrate a scenario in which the constraint qualification fails. The solution to the problem is clearly $x^* = y^* = 0$. However, is not a critical

point of the Lagrangian as there is no solution to the system of FOC

$$1 - 3\lambda x^2 = 0; \quad 2\lambda y = 0; \quad x^3 + y^2 = 0.$$

In particular, we cannot use the Lagrange Theorem to find the solution to the constraint optimization problem as $x^* = y^* = 0$ is a critical point of the constraint itself. So, MRS are not well defined.

6. Solve the following problem

$$\max_{x_1, x_2} x_1^2 x_2 \quad \text{subject to} \quad 2x_1^2 + x_2^2 = 3.$$

Solution: The constraint qualification always holds since $(0, 0)$ is the only critical point for the constraint set and never satisfies $2x_1^2 + x_2^2 = 3$. Take first order necessary conditions and observe that if $\mu \neq 0$:

$$\left. \begin{aligned} 2x_1 x_2 - 4\mu x_1 &= 0 \\ x_1^2 - 2\mu x_2 &= 0 \end{aligned} \right\} \Rightarrow x_1 = \pm x_2$$

$$2x_1^2 + x_2^2 = 3 \Rightarrow x_1^* = x_2^* = \pm 1 \text{ and } \mu^* = \pm 1/2.$$

If $\mu = 0$, a further critical point for the Lagrangian exist namely, $x_1^* = \mu^* = 0$ and $x_2^* = \pm 3^{1/2}$. As the constraint set is compact, maxima and a minima exist. We can find these by comparing the value of the objective function at all critical points. Clearly, $(1, 1)$ and $(-1, 1)$ are two global maxima. For more detail, see Simon and Blume page 418.

7. Solve the following problem when $a \in [\frac{1}{2}, \frac{3}{2}]$

$$\max_{x, y \geq 0} x^2 + y^2 \quad \text{subject to} \quad ax + y = 1.$$

Solution: In this problem both the objective function and the Lagrangian are convex. This means that we cannot use first order conditions, since these will identify a minimum and not maximum of the objective function (see also lecture 3). Hence, graphically, one can see that this problem results in a corner solution in which we choose either $(\bar{x}, \bar{y}) = (1/a, 0)$ or $(\check{x}, \check{y}) = (0, 1)$. When $a \in [1/2, 1]$, then $\bar{x}^2 = 1/a^2 > 1 = \check{y}^2$ and choosing $(\bar{x}, \bar{y})$ is optimal. On the other hand, when $a \in [1, 3/2]$ the converse inequality holds and choosing $(\check{x}, \check{y})$ is optimal. Alternatively, one can still use the Lagrange approach and find the set of critical points which satisfy the relevant conditions (including complementary slackness). One can then compute which one of these critical points maximizes the Lagrangian and thus, solve the constrained optimization problem.

8. Consider the following problem

$$\max_{x \in X} f(x) \quad \text{subject to} \quad g(x) \leq a.$$

Let X be a convex subset of $\mathbb{R}^n$, $f : X \rightarrow \mathbb{R}$ a concave function, $g : X \rightarrow \mathbb{R}^m$ a convex function, a is a vector in $\mathbb{R}^m$. What is the Lagrangian for this problem? Prove that the Lagrangian is a concave function of the choice variable x on X .

Solution: The Lagrangian for this problem satisfies:

$$L(x, \mu|a) = f(x) - \mu(g(x) - a).$$

We show that it must be concave in x . Take any two x and x' in X and observe that the restrictions in the problem require

$$\begin{aligned} f(tx + (1-t)x') &\geq tf(x) + (1-t)f(x') \Rightarrow \\ -g(tx + (1-t)x') &\geq -tg(x) - (1-t)g(x'). \end{aligned}$$

Thus for any positive vector μ , we have

$$\begin{aligned} L(tx + (1-t)x', \mu|a) &= f(tx + (1-t)x') + \mu(a - g(tx + (1-t)x')) \\ &\geq t[f(x) - \mu(g(x) - a)] + (1-t)[f(x') - \mu(g(x') - a)] \\ &= tL(x, \mu|a) + (1-t)L(x', \mu|a) \end{aligned}$$

which proves that the Lagrangian is concave.

9. Consider the problem of maximizing xyz subject to $x + y + z \leq 1$, $x \geq 0$, $y \geq 0$, and $z \geq 0$. Obviously, the three latter constraints do not bind, and we can concentrate only on the first constraint, $x + y + z \leq 1$. Find the solution and the Lagrange multiplier, and show how the optimal value would change if instead the constraint was changed to $x + y + z \leq 9/10$.

Solution: The solution satisfies $x = y = z = 1/3$ and the Lagrange multiplier satisfies $\mu = 1/9$. The optimal value is $1/27$. When the constraint is tightened to $x + y + z \leq 9/10$, $x = y = z = 3/10$ and the optimal value declines to

$$(3/10)^3 = 27/1000.$$

10. Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying:

$$f(x) = \begin{cases} u(x) & \text{if } g(x) \leq 0 \\ v(x) & \text{if } g(x) \geq 0 \end{cases}.$$

Further, suppose that: (i) $u(x) = v(x)$ if $g(x) = 0$; (ii) u and v are differentiable, strictly concave, and possess maximizer in $\mathbb{R}^n$; and (iii) $g(x)$ is differentiable and strictly convex. Carefully explain how you would solve the problem of maximizing f by choosing $x \in \mathbb{R}^n$.

Solution: Suppose $g(x) < 0$ for some $x \in \mathbb{R}^n$. If so, begin by solving the problem

$$\max_{x \in X} u(x) \text{ subject to } g(x) \leq 0.$$

Such a problem satisfies all of the conditions for KKT theorem and admits a solution as u possesses maximizer in $\mathbb{R}^n$. Hence, maximizers coincide with the solution to the following system equations:

$$\begin{aligned} 0 &= Du(x) - \lambda Dg(x), \\ 0 &= \lambda g(x), \quad 0 \leq \lambda, \quad g(x) \leq 0. \end{aligned}$$

Call a solution to this problem x_u . Suppose $g(x) > 0$ for some $x \in \mathbb{R}^n$. If so, solve the problem

$$\max_{x \in X} v(x) \quad \text{subject to} \quad -g(x) \leq 0.$$

Such a problem admits a solution as v possesses maximizer in $\mathbb{R}^n$, but does not satisfy all of the conditions for KKT theorem (as $-g(x)$ is strictly concave). Nevertheless, KKT FOC remain necessary conditions and thus maxima must satisfy system equations:

$$\begin{aligned} 0 &= Dv(x) + \lambda Dg(x), \\ 0 &= \lambda g(x), \quad 0 \leq \lambda, \quad -g(x) \leq 0. \end{aligned}$$

To find a maximizer pick amongst the solutions one that maximizes $v(x)$. Call a solution to this problem x_v . If $g(x)$ negative for all $x \in \mathbb{R}^n$, a solution is x_u . If $g(x)$ positive for all $x \in \mathbb{R}^n$, a solution is x_v . If $g(x)$ can be both strictly positive and strictly negative in $\mathbb{R}^n$, then: x_u is a solution if $u(x_u) \geq v(x_v)$; and x_v is a solution if $u(x_u) \leq v(x_v)$.

EC400 – Static Optimization and Fixed Points
Problem Set 3: Solutions

1. For $a > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & ax + y \\ \text{s.t.} \quad & x^2 + (x - y)^2 \leq 1 \\ & x \geq a, y \geq 0. \end{aligned}$$

Using the Kuhn-Tucker approach, write down the necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian? Solve the constrained optimization problem in terms of a . Find the slope of value function with respect to a , possibly relying on the Envelope Theorem.

Solutions: The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= a - \lambda_1(4x - 2y) + \lambda_2, \\ 0 &= 1 - \lambda_1(2y - 2x) + \lambda_3, \\ 0 &= \lambda_1(x^2 + (x - y)^2 - 1), \\ 0 &= \lambda_2(x - a) = \lambda_3 y, \\ 0 &\geq x^2 + (x - y)^2 - 1, \quad 0 \geq a - x, \quad 0 \geq -y, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

If $a > 1$ the constraint set is empty. Slater condition in this setting requires $a < 1$, as for any $x \geq 1$

$$x^2 + (x - y)^2 \geq x^2 \geq 1.$$

Any solutions to these conditions is maximizer of the Lagrangian as the Lagrangian is a sum of concave functions. Consider the various supports:

- At an optimum it also cannot be that $\lambda_1 = 0$, as the FOC for good x would imply that, $\lambda_2 = -a < 0$. So there is no solution on these supports.
- At an optimum it also cannot be that $\lambda_3 > 0$, as the FOC for good y would imply that

$$\lambda_3(a) = -(1 + 2\lambda_1(a)x(a)) \leq -1 < 0,$$

by $\lambda_1(a) \geq 0$ and $x(a) \geq 0$. So there can be no solution on these supports.

- At an optimum, it can be that $\lambda_1, \lambda_2 > 0$ while $\lambda_3 = 0$. If so, the two constraints would imply that

$$x(a) = a \quad \text{and} \quad y(a) = a + (1 - a^2)^{1/2}.$$

For the solution to be well defined $a < 1$ which we imposed via Slater condition. If so, the FOC for good y would imply that

$$\lambda_1(a) = \frac{1}{2(1-a^2)^{1/2}} > 0.$$

If so,

$$\begin{aligned}\lambda_2(a) &= \lambda_1(a)(4x - 2y) - a = \frac{a}{(1-a^2)^{1/2}} - 1 - a \geq 0 \\ \Leftrightarrow a^4 + 2a^3 + a^2 &\geq 2a + 1,\end{aligned}$$

which is equivalent to $a \geq 0.8832$. Thus, when a is sufficiently high this must be the solution.

- In the remaining case, the optimum must satisfy $\lambda_1 > 0$ and $\lambda_3 = \lambda_2 = 0$. Solving FOC at such values one finds that:

$$x(a) = \frac{1+a}{((1+a)^2 + 1)^{1/2}}, \quad y(a) = \frac{(2+a)}{(1+a)}x(a) > 0, \quad \lambda_1(a) = \frac{1+a}{2x(a)} > 0.$$

For this to be a solution it must be that

$$x(a) \geq a \Leftrightarrow a^4 + 2a^3 + a^2 \leq 2a + 1.$$

Thus, as this is the only point satisfying all the necessary conditions, it is the unique solution for all $a \leq 0.8832$.

The Envelope Theorem in general states that if $(\mathbf{x}(a), \boldsymbol{\lambda}(a))$ are differentiable functions and that $\mathbf{x}(a)$ satisfies the constraint qualification. If so,

$$\frac{dv(a)}{da} = \frac{\partial L(\mathbf{x}(a), \boldsymbol{\lambda}(a)|a)}{\partial a} = x(a) - \lambda_2(a)$$

If $\lambda_2(a) = 0$, to derive the result without relying on the theorem, denoting by f the objective function and by g the first constraint, observe that

$$\begin{aligned}\frac{dv(a)}{da} &= \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial a} \right] + \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right]^T \frac{\partial \mathbf{x}(a)}{\partial a} = \\ &= \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial a} \right] + \lambda_1(a) \left[\frac{\partial g(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right]^T \frac{\partial \mathbf{x}(a)}{\partial a} = \\ &= \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial a} \right] - \lambda_1(a) \frac{\partial g(\mathbf{x}(a)|a)}{\partial a} = \frac{\partial L(\mathbf{x}(a), \boldsymbol{\lambda}(a)|a)}{\partial a} = \\ &= x(a) = \frac{1+a}{((1+a)^2 + 1)^{1/2}},\end{aligned}$$

where the first equality is definitional, the second exploits FOC

$$\frac{\partial f(\mathbf{x}(a)|a)}{\partial \mathbf{x}} = \lambda_1(a) \left[\frac{\partial g(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right],$$

while the third exploits the binding constraint by totally differentiating it

$$\left[\frac{\partial g(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right]^T \frac{\partial \mathbf{x}(a)}{\partial a} = - \frac{\partial g(\mathbf{x}(a)|a)}{\partial a}.$$

Thus, the value function increases as a increases, for all $a \leq 0.8832$. If instead $a \in (0.8832, 1)$ and $\lambda_2(a) > 0$, a similar logic establishes that

$$\frac{dv(a)}{da} = x(a) - \lambda_2(a) = 2a + 1 - \frac{a}{(1 - a^2)^{1/2}}.$$

2. Let a , x and y be non-negative. Consider the problem of maximizing xy subject to $x^2 + ay^2 \leq 1$. What happens to the optimal value when a increases marginally?

Solution: The Lagrangian for the problem is $L(x, y, \mu|a) = xy + \mu(1 - x^2 - ay^2) + \mu_1 x + \mu_2 y$. Note however, that it cannot be the case that the maximizer will have either $x = 0$ or $y = 0$, since the value of the program would be zero. Thus, the solution satisfies:

$$\mu(a) = \frac{1}{2\sqrt{a}}, \quad x(a) = \frac{1}{\sqrt{2}}, \quad \text{and} \quad y(a) = \frac{1}{\sqrt{2a}}.$$

The Envelope Theorem tells us that when a increases marginally the optimal value of f declines according to

$$\frac{dx(a)y(a)}{da} = \frac{\partial L}{\partial a}(x(a), y(a), \mu(a)|a) = -\mu(a)y(a)^2 = -\frac{1}{4a^{3/2}}.$$

3. Assume that the utility function of a consumer satisfies

$$u(x, y) = x + y^{1/2}.$$

The consumer has a positive income $I > 0$ and faces positive prices $p_x > 0$ and $p_y > 0$. The consumer cannot buy negative amounts of any of the goods.

- Use Kuhn-Tucker to solve the consumer's problem.
- Show how the utility at the maximum depends on I .
- For the case of an interior solution, find how the endogenous variables change when I and p_x change. That is, compute:

$$\frac{\partial x}{\partial I}, \quad \frac{\partial y}{\partial I}, \quad \frac{\partial \mu_0}{\partial I}, \quad \frac{\partial x}{\partial p_x}, \quad \frac{\partial y}{\partial p_x}, \quad \frac{\partial \mu_0}{\partial p_x}.$$

Solution: (a) This is a quasi-linear utility function which is heavily used in economic theory. The Lagrangian is

$$L(x, y, \mu|I) = x + \sqrt{y} + \mu_0(I - p_x x - p_y y) + \mu_1 x + \mu_2 y.$$

The first order necessary conditions are

$$\begin{aligned} 1 - \mu_0 p_x + \mu_1 &= 0, \\ \frac{1}{2\sqrt{y}} - \mu_0 p_y + \mu_2 &= 0. \end{aligned}$$

Obviously the budget constraint must bind. Hence, $\mu_0 > 0$. Then consider solutions in which $\mu_1 = \mu_2 = 0$, and find the interior solution by solving FOCs for this case

$$2\sqrt{y} = \frac{p_x}{p_y} \Rightarrow y = \frac{p_x^2}{4p_y^2}, \quad x = \frac{I}{p_x} - \frac{p_x}{4p_y}, \quad \mu_0 = \frac{1}{p_x}.$$

This solution exists when $4p_y I > p_x^2$.

Now consider solutions in which $\mu_1 > 0$ and $\mu_2 = 0$. If so, FOCs imply that

$$x = 0, \quad y = \frac{I}{p_y}, \quad \mu_0 = \frac{1}{2\sqrt{I p_y}} \quad \text{and} \quad \mu_1 = \frac{1}{2\sqrt{I p_y}} p_x - 1 > 0.$$

This solution exists when $p_x > 2\sqrt{I p_y}$ or when $p_x^2 > 4p_y I$.

These two solutions exhaust all possibilities, since no solution exists in which $\mu_2 > 0$ and $\mu_1 = 0$ since $\frac{\partial u}{\partial y}(x, 0) = \infty$.

(b) First consider the solution in which $\mu_1 = \mu_2 = 0$ (ie both goods are bought). If so,

$$y^* = \frac{p_x^2}{4p_y^2} \quad \text{and} \quad x^* = \frac{I}{p_x} - \frac{p_x}{4p_y}.$$

and hence

$$u(x^*, y^*) = \frac{I}{p_x} - \frac{p_x}{4p_y} + \frac{p_x}{2p_y}.$$

By taking the derivative with respect to I , we get

$$\frac{\partial u(x^*, y^*)}{\partial I} = \frac{1}{p_x} = \mu_0.$$

Finally, consider a corner solution, for example when $\mu_1 > 0$. Then, $x^* = 0$ and $y^* = \frac{I}{p_y}$ and hence

$$u(x^*, y^*) = \sqrt{\frac{I}{p_y}} \Rightarrow \frac{\partial u(x^*, y^*)}{\partial I} = \frac{1}{2\sqrt{p_y I}} = \mu_0.$$

(c) Since $\mu_1 = \mu_2 = 0$, the first conditions of this problem reduce to:

$$\begin{aligned} l - p_x x - p_y y &= 0, \\ 1 - \mu_0 p_x &= 0, \\ \frac{1}{2\sqrt{y}} - \mu_0 p_y &= 0. \end{aligned}$$

Consider the determinant of the Jacobian of this system of three simultaneous equations,

$$\begin{vmatrix} 0 & -p_x & -p_y \\ -p_x & 0 & 0 \\ -p_y & 0 & -\frac{1}{4y^{3/2}} \end{vmatrix} = p_x^2 \frac{1}{4y^{3/2}} > 0.$$

This is not surprising, since the original Lagrangian is concave. At the unique interior optimum, $y > 0$, the bordered Hessian, H , is negative definite (satisfying second order conditions – ie the largest $n - e = 1$ leading minors has the same sign as $(-1)^n = (-1)^2$), since

$$|H_3| > 0.$$

Using total differentiation with respect to l , and assuming an interior solution (ie $\mu_1 = \mu_2 = 0$), we have

$$\begin{aligned} dl - p_x dx - p_y dy &= 0, \\ -p_x d\mu_0 &= 0, \\ -\frac{1}{4y^{3/2}} dy - p_y d\mu_0 &= 0. \end{aligned}$$

It is obvious that $d\mu_0 = 0$ and hence $dy = 0$. Thus,

$$\frac{\partial x}{\partial l} = \frac{1}{p_x}, \quad \frac{\partial y}{\partial l} = \frac{\partial \mu_0}{\partial l} = 0.$$

Now, consider a change in p_x

$$\begin{aligned} -p_x dx - x dp_x - p_y dy &= 0, \\ -p_x d\mu_0 - \mu_0 dp_x &= 0, \\ -\frac{1}{4y^{3/2}} dy - p_y d\mu_0 &= 0, \end{aligned}$$

or in matrix notation

$$\begin{pmatrix} 0 & -p_x & -p_y \\ -p_x & 0 & 0 \\ -p_y & 0 & -\frac{1}{4y^{3/2}} \end{pmatrix} \begin{pmatrix} \frac{\partial \mu_0}{\partial p_x} \\ \frac{\partial x}{\partial p_x} \\ \frac{\partial y}{\partial p_x} \end{pmatrix} = \begin{pmatrix} x \\ \mu_0 \\ 0 \end{pmatrix}$$

and hence one can use Cramer's rule to solve this system, since

$$\begin{vmatrix} 0 & -p_x & -p_y \\ -p_x & 0 & 0 \\ -p_y & 0 & -\frac{1}{4y^{3/2}} \end{vmatrix} > 0.$$

Therefore, we have that (verify please)

$$\begin{aligned} \frac{\partial \mu_0}{\partial p_x} &= -\frac{\mu_0}{p_x} < 0, \\ \frac{\partial y}{\partial p_x} &= 4\frac{p_y}{p_x}\mu_0 y^{3/2} > 0, \\ \frac{\partial x}{\partial p_x} &= -\frac{x}{p_x} - 4\frac{p_y^2}{p_x^2}\mu_0 y^{3/2} < 0. \end{aligned}$$

4. Consider the system of equations:

$$\mathbf{F}(\mathbf{x}|b) = \begin{cases} x_1 - e^{x_2} \\ e^{x_1} + x_2 - b^2 \end{cases} = 0.$$

Argue whether you can invoke the implicit function theorem. Then, exploit the theorem where possible to find the slopes of the implicit functions $\mathbf{f}(b)$.

Solution: Observe that

$$|D_{\mathbf{x}}\mathbf{F}(\mathbf{x}|b)| = 1 + e^{x_1+x_2} > 0.$$

Thus, IFT applies and there exists implicit functions such that

$$\mathbf{F}(\mathbf{f}(b)|b) = \begin{cases} f_1(b) - e^{f_2(b)} \\ e^{f_1(b)} + f_2(b) - b^2 \end{cases} = 0.$$

By totally differentiating the system, we find

$$\frac{d\mathbf{F}(\mathbf{f}(b)|b)}{db} = \begin{cases} f_1'(b) - f_2'(b)e^{f_2(b)} \\ f_1'(b)e^{f_1(b)} + f_2'(b) - 2b \end{cases} = 0.$$

Consequently, by solving the linear system and requiring $\mathbf{x} = \mathbf{f}(b)$, we find that

$$\frac{\partial \mathbf{f}(b)}{\partial b} = - \left[\frac{\partial \mathbf{F}(\mathbf{f}(b)|b)}{\partial \mathbf{x}} \right]^{-1} \frac{\partial \mathbf{F}(\mathbf{f}(b)|b)}{\partial b} = \begin{cases} 2be^{x_2}/(1 + e^{x_1+x_2}) \\ 2b/(1 + e^{x_1+x_2}) \end{cases}.$$

5. For each of the following correspondences $h : \mathbb{R} \rightrightarrows \mathbb{R}$ show whether they are convex-valued, upper-hemicontinuous or lower-hemicontinuous:

(a) $h(x) = [5x, 10x]$ for $x \in [0, 1]$;

(b) $h(x) = \{5x, 10x\}$ for $x \in [0, 1]$;

$$(c) \ h(x) = \begin{cases} 1/2 & \text{for } x \in [0, 1) \\ [0, 1] & \text{for } x = 1 \end{cases}.$$

Solution: (a) Not UHC, as the graph is open; convex-valued as any interval $[5x, 10x)$ is convex; LHC as any $(y, x) \in Gr(h)$ and for any sequence $\{x^k\}_{k=1}^{\infty}$ such that $x^k \rightarrow x$, there exists a sequence $\{y^k\}_{k=1}^{\infty}$ such that

$$y^k \in [5x^k, 10x^k) \text{ for all } k \text{ and } y^k \rightarrow y.$$

(b) UHC and LHC obviously as graph is closed and union of two continuous functions; not convex-valued obviously; (c) UHC as graph is closed; obviously convex-valued; not LHC as $(1, 1) \in Gr(h)$, but for any sequence $\{x^k\}_{k=1}^{\infty}$ such that $x^k \rightarrow 1$, we have that

$$y^k \in h(x^k) \text{ for all } k \text{ implies } y^k \rightarrow 1/2.$$

6. Consider the correspondence $h : \mathbb{R} \rightrightarrows \mathbb{R}$ where

$$h(x) = \begin{cases} [0, 1/2] & \text{if } x > 1/3 \\ [1/2, 1] & \text{if } x < 1/3 \end{cases}.$$

Argue whether the Kakutani's Fixed Point Theorem applies and find all the fixed points.

Solution: The correspondence is empty at $1/3$. Thus, the Fixed Point Theorem does not apply. However, any $x \in (1/3, 1/2]$ is a fixed point. Obviously there are no fixed points for $x < 1/3$ as $h(x) > 1/2$ and similarly there are no fixed points for $x > 1/2$ as $h(x) < 1/2$.

7. Consider the correspondence $h : \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$,

$$h_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_2 > 1/3 \\ [0, 1] & \text{if } x_2 = 1/3 \\ 0 & \text{if } x_2 < 1/3 \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} 1 & \text{if } x_1 < 1/2 \\ [0, 1] & \text{if } x_1 = 1/2 \\ 0 & \text{if } x_1 > 1/2 \end{cases}.$$

Show how you can exploit Kakutani's Fixed Point Theorem despite the domain being open, and find all the fixed points.

Solution: Both correspondences have closed graph and are therefore UHC. Moreover, both are obviously convex-valued and non-empty. Since the image of $\mathbf{h}$ belongs to $[0, 1]^2$, the Fixed Point Theorem applies when you restrict the domain to $[0, 1]^2$. Thus, there will be at least one fixed point on $[0, 1]^2$, and there can be no fixed points outside $[0, 1]^2$ because there is no image outside $[0, 1]^2$. Furthermore, there is only one fixed point in which $\mathbf{x} = (1/2, 1/3)$.

8. For all values of $b \geq 0$, solve the problem

$$\mathbf{x}^*(b) = \arg \max_{\mathbf{x} \in [0,1]^2} bx_1 + x_2 \quad \text{s.t.} \quad x_1 + x_2 = 1.$$

Show that $\mathbf{x}^*(b)$ is non-empty, compact-valued and convex-valued. Would this have followed also from the theorem of the maximum? Are the assumptions of the theorem satisfied?

Solution: Solving the maximization problem yields

$$\mathbf{x}(b) = \begin{cases} (1, 0) & \text{if } b > 1 \\ \{(p, 1-p) \mid p \in [0, 1]\} & \text{if } b = 1 \\ (0, 1) & \text{if } b < 1 \end{cases}$$

This correspondence is clearly non-empty, compact-valued and convex-valued. We could have obtained the same insight from the Theorem of the Maximum as $[0, 1]^2$ is compact and as $f(\mathbf{x}|b) = bx_1 + x_2$ is a continuous function, and as $g(b) = \{\mathbf{x} \mid x_1 + x_2 = 1\}$ is a continuous correspondence in b .